RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
# INDUSTRIAL ENGINEERING & MANAGEMENT

| | | |
|---|---|---|
| **Date : 17th June 2026** | **Improvement CIE** | **Max. Marks : 10 + 50** |
| **Semester : VI** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: Entrepreneurship and Intellectual Property Rights** | | **Course Code : HS361TA** |

Note:
1. Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | What is the acquisition of a company? | 2 | 1 | 3 |
| 2. | Define Unique Value Proposition (UVP). | 2 | 2 | 4 |
| 3. | What is market segmentation? | 2 | 1 | 4 |
| 4. | What is the primary purpose of market analysis in a business plan? | 2 | 2 | 4 |
| 5. | What is crowdfunding? | 2 | 2 | 4 |
| | **Part – B** | | | |
| 1 | A group of engineering graduates plans to start a company manufacturing biodegradable packaging materials. They have identified increasing demand from e-commerce companies due to environmental regulations. However, several established plastic packaging manufacturers dominate the market. The founders need to prepare a business plan and develop a strategy to compete effectively.<br>**Questions:**<br>a) Prepare the key components of the business plan for this startup.<br>b) Suggest an appropriate competitive strategy and justify your choice. | 10 | 3 | 3 |
| 2 | A successful restaurant chain operating in Bengaluru wants to expand into neighboring states. The management is debating whether to grow through opening new outlets independently or by partnering with existing regional restaurants.<br>**Questions:**<br>a) Conduct a SWOC analysis for the restaurant chain.<br>b) Recommend an appropriate growth strategy. | 10 | 3 | 4 |
| 3 | Discuss the importance of Customer Relationship Management (CRM) in entrepreneurship. | 10 | 2 | 4 |
| 4 | A startup selling handmade eco-friendly crafts has launched an online store. Despite having quality products, sales remain low due to limited online visibility and poor customer engagement. The entrepreneur has a limited marketing budget and wants to increase brand awareness and sales.<br>**Questions:**<br>a) Explain how the entrepreneur can use the 4Ps of Marketing to improve sales.<br>b) Recommend suitable digital marketing strategies. | 10 | 1 | 4 |

| | | | | |
|---|---|---|---|---|
| 5 | A startup has grown from 10 to 80 employees within two years. However, employee turnover has increased and productivity has declined. The founder believes that human resource practices need improvement.<br>**Questions:**<br>a) Identify the HR issues faced by the startup.<br>b) Recommend appropriate HR management practices. | 10 | 1 | 4 |

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Department of Civil Engineering

| | | | |
|---|---|---|---|
| **Date** | **12/01/2026** | **Maximum Marks** | **10 + 50** |
| **Course Code** | **HS351TA** | **Duration** | **20 Min + 90 Min** |
| **Sem** | **V** | **Improvement CIE** | |
| **ENTREPRENEURSHIP & INTELLECTUAL PROPERTY RIGHTS** | | | |

**Part A**

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| 1. | Define Entrepreneurial Marketing and state one distinguishing feature that differentiates it from traditional marketing. | 2 | 1 | 3 |
| 2. | Identify any two bases of market segmentation used in the STP framework. | 2 | 1 | 3 |
| 3. | List any two sources of entrepreneurial financing available to start-ups. | 2 | 2 | 3 |
| 4. | Define an Executive Summary and state any two key purposes it serves in a Business Plan. | 2 | 3 | 3 |
| 5. | Identify the Porter's Generic Strategy adopted by a firm that targets a niche market with customized products, and justify your answer briefly. | 2 | 2 | 3 |

**Part B**

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| 1. | Analyze the key elements of a comprehensive Business Plan. Explain the role and significance of the Executive Summary, Company Description, and Market Analysis in translating a business idea into a viable entrepreneurial venture. | 10 | 2 | 2 |
| 2. | Analyse how the 4Ps of Marketing (Product, Price, Place, Promotion) work together in an entrepreneurial context. Illustrate your answer with one Indian start-up case example, highlighting how integration of the marketing mix creates competitive advantage. | 10 | 3 | 3 |
| 3. | Evaluate the role of Market Segmentation, Targeting, and Positioning (STP) in developing an effective entrepreneurial marketing strategy. Explain how STP is integrated with the marketing mix, using a real-world business illustration. | 10 | 2 | 3 |
| 4. | Explain the major sources of entrepreneurial financing and compare equity financing and debt financing with respect to control, risk, and long-term sustainability of a start-up. Support your answer with suitable examples. | 10 | 3 | 4 |
| 5. | Evaluate the importance of Strategic Planning in business development. Discuss how Vision, Mission, Goals, and Objectives, supported by SWOC Analysis, influence the selection of competitive strategies such as Cost Leadership, Differentiation, and Focus Strategy, and enable sustainable growth through Organic Growth, Mergers & Acquisitions, or Strategic Alliances. | 10 | 3 | 3 |

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DEPARTMENT OF
# INDUSTRIAL ENGINEERING & MANAGEMENT

| | | |
|---|---|---|
| **Date : 23rd June 2025** | **Improvement CIE** | **Max. Marks : 10 + 50** |
| **Semester : VI** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: Entrepreneurship and Intellectual Property Rights** | | **Course Code : HS361TA** |

Note:
1. Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | List 4Ps of marketing? | 2 | 1 | 3 |
| 2. | Define Unique Value Proposition (UVP)? | 2 | 2 | 4 |
| 3. | What is budgeting? | 2 | 1 | 4 |
| 4. | What is 'fair use' in copyright law? | 2 | 2 | 4 |
| 5. | List any two rights granted to broadcasting organizations and performers? | 2 | 2 | 4 |
| | **Part – B** | | | |
| 1 | Explain branding with examples. How do Branding contribute to business success? | 10 | 2 | 3 |
| 2 | Describe the key functions of Human Resource Management. | 10 | 2 | 4 |
| 3 | A startup company designs an ergonomic and uniquely shaped water bottle with an aesthetic twist for sports users. They want to protect the design under Indian law. Explain the features of industrial design protection and the procedure to obtain protection. | 10 | 2 | 4 |
| 4 | A freelance photographer named Arjun took a series of wildlife photographs during a jungle safari. One of his photos went viral after being posted on social media. A popular travel magazine downloaded the image and used it in a commercial advertisement without seeking Arjun's permission or providing any credit. Arjun discovers this usage and wants to take legal action.<br>Discuss the rights Arjun has under the Copyright Act, 1957, possible remedies available to him. | 10 | 1 | 4 |

| | | | | |
|---|---|---|---|---|
| 5 | InnoTech Solutions, a Bangalore-based AI startup, developed a proprietary algorithm for real-time fraud detection in digital payments. The startup had not patented the algorithm to maintain it as a trade secret. A junior developer, Akhil, who worked on the codebase, left and started his own company using a similar algorithm.<br>InnoTech sued Akhil for misappropriation of trade secrets, citing his employment contract's confidentiality clause.<br>a) Discuss the legal provisions or frameworks applicable for protecting trade secrets in India. Which laws or doctrines can be used in the absence of a dedicated Trade Secrets Act? Evidence Required<br>b) Analyze the possible legal arguments both the plaintiff and the defendant can present in a trade secret misappropriation case<br>c) What type of evidence is essential to prove trade secret infringement or misuse in court? Provide examples relevant to the case. | 10 | 1 | 4 |

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## Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
## INDUSTRIAL ENGINEERING & MANAGEMENT

| | | | |
|---|---|---|---|
| **Date** | **27th January 2025** | **Maximum Marks** | **10 + 50** |
| **Course Code** | **HS351TA** | **Duration** | **120 Mins** |
| **Sem** | **V Semester** | **Improvement CIE** | |
| **ENTREPRENEURSHIP AND INTELLECTUAL PROPERTY RIGHTS** | | | |

Note:
1. Answer all the Questions.

**PART – A**

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| 1 | _____ is a fundamental framework in marketing that helps businesses identify and reach their most valuable customers. | 1 | 1 | 2 |
| 2 | Creating a _____ is about identifying and clearly communicating what makes a product, service, or brand distinctively valuable to its target audience | 1 | 1 | 4 |
| 3 | A company is launching a premium product with a production cost of Rs 30 per unit. They estimate fixed costs for branding and marketing at Rs 2,00,000. The company plans to sell 10,000 units in the first year and targets a 40% profit margin. What should be the selling price per unit to achieve the desired profit margin? | 2 | 2 | 2 |
| 4 | Toyota developed the **Toyota Way**, a set of principles that focuses on _____ and respect for people. | 1 | 1 | 4 |
| 5 | Differentiate between Planning and Strategy. | 2 | 1 | 2 |
| 6 | With _____ growth, a company expands through its own operations using its own internal resources. | 1 | 1 | 4 |
| 7 | If a company's gross profit margin is increasing but its net profit margin is declining, what could be the reason? | 2 | 2 | 3 |

**PART – B**

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| 1. | A tech company is developing a mobile app that combines fitness tracking with AI-based personalized coaching. They are seeking funding from venture capitalists. How can the company effectively articulate its unique value proposition in the business plan to stand out from competitors in the crowded fitness tech market? | 10 | 4 | 4 |

## Academic year 2024-2025 (ODD Sem)

| | | | | |
|---|---|---|---|---|
| 2. | a) A start-up has developed a patent-pending water purification device that uses renewable energy. They aim to market it in developing countries. How should they address intellectual property, innovation, and scalability in their business plan to appeal to investors?<br>b) A global e-commerce platform is considering a strategic alliance with a logistics company to enhance delivery efficiency in underserved regions. The e-commerce platform wants to improve delivery times, and the logistics company seeks access to a broader customer base. What factors should both companies consider when forming a strategic alliance, and how can they ensure that the partnership is mutually beneficial? | 5<br><br>5 | 4<br><br>3 | 4<br><br>4 |
| 3. | What are the objectives of Financial Management? Investigate Financial Management Tools and Techniques. | 10 | 2 | 2 |
| 4. | Illustrate how successful companies have leveraged each of the 4Ps to achieve market success. | 10 | 3 | 2 |
| 5. | A manufacturing company needs Rs 20 Lacs to purchase advanced machinery. They are considering two options:<br>• Taking a bank loan with a 7% annual interest rate.<br>• Selling 20% of the company's equity to a private investor.<br>1. What factors should the company consider when choosing between equity and debt financing? Which option might be better if the company expects rapid growth? Justify with a case study.<br>2. What steps can the company take to manage debt repayment and avoid defaulting on the loan?<br>3. Illustrate how successful companies have leveraged each of the 4Ps to achieve market success. | 10 | 4 | 2 |

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

| **Particulars** | | | CO1 | CO2 | CO3 | CO4 | CO5 | L1 | L2 | L3 | L4 | L5 | L6 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Marks Distribution** | Max Marks | Quiz | – | 5 | 2 | 3 | – | 06 | 04 | – | – | – | – |
| | | Test | – | 20 | 10 | 20 | – | – | 10 | 15 | 25 | – | – |

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DEPARTMENT OF
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|---|---|---|
| **Date : 12/01/2026** | **Improvement CIE** | **Max. Marks : 10 + 50** |
| **Semester : V** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: Operations Management** | | **Course Code : IM352IA** |

Note:
1. Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | A _____ shop deals with customized, low-volume production. | 1 | 1 | 2 |
| 2. | In mass production, the layout used is _____ layout. | 1 | 1 | 2 |
| 3. | The maximum output a system can produce is called its _____. | 1 | 1 | 2 |
| 4. | The long-term decision of choosing a geographic site for operations is called _____. | 1 | 2 | 3 |
| 5. | Continuous processes are suitable for _____ production. | 1 | 2 | 3 |
| 6. | A bottleneck is a stage that limits the _____ of the system. | 1 | 2 | 3 |
| 7. | The Theory of _____ focuses on improving the weakest link in a process. | 1 | 2 | 3 |
| 8. | MRP stands for _____ Requirements Planning. | 1 | 1 | 3 |
| 9. | The detailed schedule of what is to be produced is called the _____. | 1 | 2 | 3 |
| 10. | ERP stands for _____ Resource Planning. | 1 | 1 | 4 |
| | **Part – B** | | | |
| 1 | Explain the concepts of economies and diseconomies of scale with appropriate examples. | 10 | 2 | 2 |
| 2 | List and explain the three major inputs for a Material Requirement Planning system. | 10 | 2 | 2 |
| 3 | Explain process reengineering and process improvement, highlighting their differences with suitable examples. | 10 | 4 | 3 |
| 4 | Describe a systematic approach to process analysis with the help of a block diagram. | 10 | 2 | 2 |

<table>
<tr>
<td>5</td>
<td>A copy center in an office building prepares bound reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to run, collate, and bind each copy depends on, among other factors, the number of pages. The center operates 250 days per year, with one 8-hour shift. Management believes that a capacity cushion of 15 percent (beyond the allowance built into time standards) is best. It currently has three copy machines. Based on the following table of information, determine how many machines are needed at the copy center.
<table>
<tr><th>Item</th><th>Client X</th><th>Client Y</th></tr>
<tr><td>Annual demand forecast (copies)</td><td>2,000</td><td>6,000</td></tr>
<tr><td>Standard processing time (hour/copy)</td><td>0.5</td><td>0.7</td></tr>
<tr><td>Average lot size (copies per report)</td><td>20</td><td>30</td></tr>
<tr><td>Standard setup time (hours)</td><td>0.25</td><td>0.40</td></tr>
</table>
</td>
<td>10</td>
<td>3</td>
<td>3</td>
</tr>
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## Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
## INDUSTRIAL ENGINEERING & MANAGEMENT

| | | | |
|---|---|---|---|
| **Date** | **27/01/2025** | **Maximum Marks** | **10 + 50** |
| **Course Code** | **IM352IA** | **Duration** | **30 + 90 Min** |
| **Sem** | **V** | **CIE – III** | |
| **OPERATIONS MANAGEMENT** | | | |

Note:
1. Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1 | A _____ is any resource whose capacity is less than or equal to the demand placed on it. | 1 | 1 | 3 |
| 2 | In product mix decisions, TOC prioritizes products based on _____ per unit of the bottleneck resource. | 1 | 1 | 3 |
| 3 | The Theory of Constraints uses the _____ process to repeat improvements once a bottleneck is resolved. | 1 | 1 | 3 |
| 4 | In manufacturing, reducing _____ time at a bottleneck can significantly improve efficiency. | 1 | 1 | 3 |
| 5 | Differentiate between Static Schedule and Dynamic Schedule | 2 | 2 | 4 |
| 6 | What is bottleneck scheduling, and why is it critical? | 2 | 2 | 4 |
| 7 | Using the Earliest Due Date (EDD) rule, determine the sequence for the following jobs based on their due dates:<br>Job / Processing time(hr) / Due date(hr)<br>A / 4 / 10<br>B / 3 / 8<br>C / 2 / 6<br>D / 5 / 12 | 2 | 2 | 4 |
| | **Part –B** | | | |
| 1 | Explain how the Theory of Constraints can be applied to manage bottlenecks in a manufacturing environment. Provide examples of how identifying and addressing bottlenecks can improve overall production efficiency. | 10 | 3 | 3 |

## Academic year 2024-2025 (ODD Sem)

| | | | | |
|---|---|---|---|---|
| 2 | Consider the following flow shop scheduling problem with two machines (Machine 1 and Machine 2) and five jobs. The processing times (in hours) for each job on the two machines are as follows:<br>Job \| Machine 1 \| Machine 2<br>A \| 5 \| 4<br>B \| 7 \| 6<br>C \| 3 \| 5<br>D \| 4 \| 3<br>E \| 6 \| 7<br>Using Johnson's Rule, find the optimal job sequence that minimizes the makespan. | 10 | 3 | 4 |
| 3 | Consider the following 3 machines and 5 jobs flow shop problem.<br>Job, $i$ \| Processing Time<br>\| Machine 1 \| Machine 2 \| Machine 3<br>1 \| 10 \| 10 \| 12<br>2 \| 12 \| 8 \| 20<br>3 \| 16 \| 6 \| 14<br>4 \| 12 \| 4 \| 20<br>5 \| 20 \| 8 \| 8<br>Check whether the Johnson's algorithm can be extended to this problem. If yes, find the optimal sequence and the corresponding makespan. | 15 | 3 | 4 |
| 4 | Use graphical method to minimize the time needed to process the following jobs on the machines shown (i.e. for each machine, find the job which should be scheduled first). Also, calculate the total time elapsed to complete both jobs.<br>Job 1 \| Sequence \| A \| B \| C \| D \| E<br>\| Time (Hrs). \| 4 \| 3 \| 6 \| 2 \| 7<br>Job 2 \| Sequence \| C \| B \| E \| D \| A<br>\| Time (Hrs) \| 6 \| 3 \| 5 \| 3 \| 7 | 15 | 4 | 4 |

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

| **Marks Distribution** | **Particulars** | | CO1 | CO2 | CO3 | CO4 | L1 | L2 | L3 | L4 | L5 | L6 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Max Marks | Quiz | – | – | 4 | 6 | 4 | 6 | – | – | – | – |
| | | Test | – | – | 10 | 40 | – | – | 35 | 15 | – | – |

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DEPARTMENT OF
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|---|---|---|
| **Date : 14/01/2026** | **Improvement CIE** | **Max. Marks : 10 + 50** |
| **Semester : V** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: Financial Accounting & Costing** | | **Course Code : IM254TA** |

Note: Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **PART - A** | | | |
| 1 | State any two objectives of standard costing. | 2 | 1 | 3 |
| 2 | List the components of standard cost. | 2 | 3 | 4 |
| 3 | Define material cost variance. Write its formula. | 2 | 3 | 4 |
| 4 | Comment on Material Price Variance with formula. | 2 | 3 | 4 |
| 5 | List the main components of a cost sheet. | 2 | 2 | 3 |
| | **PART - B** | | | |
| 1 | Calculate prime cost from the following information:<br>Opening stock of raw material = Rs. 12,500<br>Purchased raw material = Rs. 75,000<br>Expenses incurred on raw material =Rs. 5,000<br>Closing stock of raw material = Rs. 22,500<br>Wages Rs. 47,600<br>Direct expenses Rs. 23,400 | 10 | 2 | 3 |
| 2 | Super Specialities Ltd has to supply 2,40,000 units of a component "Wizzo" annually for its valued customer Goenka. The Company has been producing 20,000 units per month by having 12 runs per annum. Its new Finance Manager Arun says that the production should be brought down to 5,000 units per batch and 48 batches should be run per annum. You are required to advise the Company on the following issues given that the Set up Cost per batch is Rs. 75 per batch and Carrying Cost per unit is Rs. 1 per annum.<br>1. What is the Economic Batch Size?<br>2. Should the Company continue producing 20,000 units per batch or should it adopt Arun's suggestion?<br>3. For least cost, how many batches should be run in a year?<br>4. What should be the interval between each batch? | 10 | 4 | 3 |
| 3 | In a manufacturing unit, raw material passes through four processes, I, II, III, and IV and the output of each process is the input for the subsequent process. The losses in the four processes are respectively 25%, 20%, 20% and 16 2/3 % respectively for I, II, III and IV processes of the input. If the end product at the end of the IV process is 40,000 kg, what is the quantity of raw material required to be fed at the beginning of Process I and the cost of the same at Rs. 5 per kg? | 10 | 4 | 4 |
| 4 | The standard cost card shows the following details relating to the materials needed to … | 10 | 2 | 4 |

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## Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
## INDUSTRIAL ENGINEERING & MANAGEMENT

| | | | |
|---|---|---|---|
| **Date** | **28th January 2025** | **Maximum Marks** | **10 + 50** |
| **Course Code** | **IM254TA** | **Duration** | **120 Mins** |
| **Sem** | **V Semester** | **Improvement Test** | |
| **FINANCIAL ACCOUNTING AND COSTING** | | | |

Note:
1. Answer all the Questions.

**PART – A**

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| 1 | Mention the main purpose of a Profit and Loss Account? | 02 | 2 | CO3 |
| 2 | List the two main sections of a Balance Sheet? | 02 | 1 | CO3 |
| 3 | Differentiate between Cost of Production and Cost of Sales? | 02 | 2 | CO1 |
| 4 | What are the main components of Batch Costing? | 02 | 1 | CO4 |
| 5 | What is the meaning of 'Equivalent Units' in process costing? | 02 | 2 | CO4 |

**PART – B**

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| 1. | Prepare a Trading and Profit and Loss Account for the year ending 31st December 2024 from the following information. Also interpret the financial status of the organization. | 10 | 4 | CO3 |

| Particulars | Amount (₹) |
|---|---|
| Opening Stock | 1,00,000 |
| Purchases | 5,00,000 |
| Sales | 8,00,000 |
| Wages | 50,000 |
| Carriage Inwards | 20,000 |
| Salaries | 60,000 |
| Office Rent | 25,000 |
| Depreciation on Furniture (10%) | 15,000 |
| Interest on Loan | 10,000 |
| Closing Stock | 1,50,000 |

## Academic year 2024-2025 (ODD Sem)

| | | | | |
|---|---|---|---|---|
| 2. | Prepare a Balance Sheet for ABC & Co. as on 31st December 2024 using the following details:<br>• Capital: ₹3,00,000<br>• Profit for the Year: ₹50,000<br>• Machinery: ₹2,50,000<br>• Creditors: ₹1,00,000<br>• Cash in Hand: ₹20,000<br>• Debtors: ₹1,00,000<br>• Bank Loan: ₹50,000<br>• Stock: ₹1,30,000 | 10 | 3 | CO3 |
| 3. | Prepare a Cost Sheet for XYZ Ltd. for the year ending 31st March 2024 from the following data:<br>• Raw Material Purchased: ₹2,00,000<br>• Freight on Raw Materials: ₹20,000<br>• Direct Labor: ₹1,50,000<br>• Direct Expenses: ₹25,000<br>• Factory Overheads: ₹1,00,000 (60% of Direct Labor)<br>• Office Overheads: ₹75,000 (10% of Factory Cost)<br>• Selling Overheads: ₹50,000<br>• Distribution Overheads: ₹30,000<br>• Profit: ₹1,20,000 | 10 | 3 | CO4 |
| 4. | XYZ Ltd. is working on Job No. 45. The following details are provided for the job:<br>• Direct Materials Used: ₹50,000<br>• Direct Wages: ₹30,000<br>• Machine Hours Worked: 20 hours<br>• Machine Hour Rate: ₹500 per hour<br>• Office Overheads: 20% of Prime Cost<br>• Profit Markup: 25% on Total Cost<br>Prepare a Job Cost Sheet for Job No. 45. | 10 | 3 | CO4 |
| 5. | XYZ Ltd. uses process costing to manufacture goods that go through three processes: A, B, and C. The following information is for Process B for the month of January 2025:<br>• Opening Work-in-Progress (WIP): 500 units valued at ₹30,000 (Material: ₹20,000; Labor: ₹5,000; Overheads: ₹5,000)<br>• Raw Materials Added: ₹40,000<br>• Direct Labor: ₹25,000<br>• Factory Overheads: ₹12,500<br>• Units Introduced from Process A: 5,000<br>• Units Transferred to Process C: 4,800<br>• Closing WIP: 700 units (Material 100% complete, Conversion 50% complete)<br>Prepare the Cost Sheet for Process B. | 10 | 3 | CO4 |

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

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## Academic year 2024-2025 (ODD Sem)

| Particulars | | | CO1 | CO2 | CO3 | CO4 | CO5 | L1 | L2 | L3 | L4 | L5 | L6 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Marks Distribution** | Max Marks | Quiz | 02 | – | 04 | 04 | – | 02 | – | 04 | 04 | – | – |
| | | Test | – | – | 20 | 30 | – | – | – | – | 40 | 10 | – |

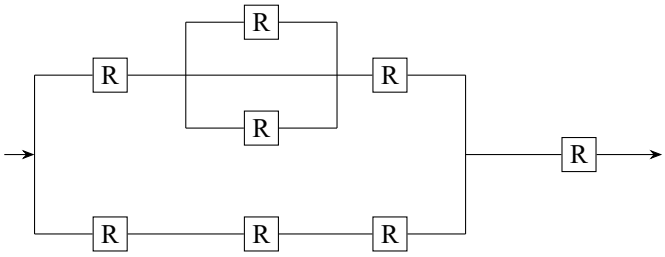
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DEPARTMENT OF
# INDUSTRIAL ENGINEERING & MANAGEMENT

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|---|---|---|
| **Date : 13th Jan 2026** | **IMPROVEMENT CIE** | **Max. Marks : 10 + 50** |
| **Semester : V** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: Quality Assurance** | | **Course Code : IM353IA** |

Note:
1. Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | State any two principles of experimentation. | 2 | 3 | 01 |
| 2. | What is a factorial experiment? | 2 | 2 | 01 |
| 3. | What is MTBF? | 2 | 2 | 02 |
| 4. | What are the three common types of failures? | 2 | 3 | 02 |
| 5. | What is a parallel system in reliability? | 2 | 2 | 02 |
| | **Part – B** | | | |
| 1 | Consider the following series-parallel configuration:<br>  <br>i) Derive the expression for system reliability in terms of the component reliabilities, stating clearly the assumption made.<br>ii) Compute the System reliability for $R = 0.9$ and $R = 0.95$. | 10 | 2 | 3 |
| 2 | Explain MTBF, MTTF, MTTR relationships with examples. | 10 | 3 | 3 |
| 3 | Derive and explain reliability for two components in parallel. | 10 | 4 | 3 |
| 4 | Explain the principles of experimentation in detail. | 10 | 4 | 4 |
| 5 | Explain six examples of designed experiments in process improvement. | 10 | 4 | 4 |

## Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
## INDUSTRIAL ENGINEERING & MANAGEMENT

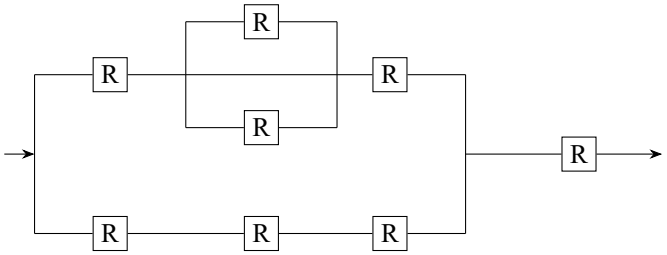
| | | | |
|---|---|---|---|
| **Date** | **28th January 2025** | **Maximum Marks** | **50** |
| **Course Code** | **IM353IA** | **Duration** | **90 Min** |
| **Sem** | **V** | **Improvement CIE** | |
| **QUALITY ASSURANCE** | | | |

Note:
1. Answer all the Questions.

**PART – A** **Max Marks: 10**

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| 1 | Three components each with a reliability of 0.9 are placed in series. What is the reliability of the system? | 2 | L2 | CO1 |
| 2 | There are two components that are identical in a parallel configuration where the second unit is in standby and will be activated only upon failure of the primary unit. The switching to the backup unit is considered certain. The failure rate for each component is 0.002 failures/hour. What is the reliability for 200 hours? | 2 | L1 | CO2 |
| 3 | The failure rates for two units in a parallel configuration, where the secondary unit is on back up until switched on upon failure of the primary unit, are each 0.002 failures/hour and the switch reliability is 0.95 per operation. What is the reliability for 300 hours? | 2 | L2 | CO3 |
| 4 | List the Advantages of CRD | 2 | L2 | CO3 |
| 5 | Expand ANOVA and ANCOVA | 2 | L2 | CO2 |

**PART – B** **Max Marks: 50**

| Sl. No | Questions | M | CO | BT |
|---|---|---|---|---|
| 1 | Define the following terms: Reliability; MTTF; MTBF; MTTR; Availability ,Maintainability | 10 | CO3 | 3 |
| 2 | Explain the utility of 'bath tub curve'. What are the reasons for failure during the three stages of equipment life? | 10 | CO1 | 3 |
| 3 | Consider the following series-parallel configuration:<br>  <br><br>Compute the System reliability for $R = 0.96$. | 10 | CO2 | 3 |
| 4 | List the General practical steps and guidelines for planning and conducting DOE | 10 | CO4 | 4 |

## Academic year 2024-2025 (ODD Sem)

| | | | | |
|---|---|---|---|---|
| 5.a | How do the principles of replication and randomization work together to strengthen the internal validity of an experiment? | 5 | CO2 | 3 |
| 5.b | What are the key assumptions behind the application of a full factorial design and how do they affect the interpretation of experimental results? | 5 | CO2 | 3 |

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

| **Marks Distribution** | **Particulars** | | CO1 | CO2 | CO3 | CO4 | CO5 | L1 | L2 | L3 | L4 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Max Marks | Quiz | 02 | 04 | 04 | – | – | 02 | 08 | – | – |
| | | Test | 10 | 20 | 10 | 10 | – | – | 30 | 10 | 10 |

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